Information Engineering“IE —INT 461

3–Data and Process Modelling

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Outline

1. Chapter Preview and objectives
2. Guidelines for Drawing DFDs
   - The naming convention
3. DATA DICTIONARY
4. Summery of the chapter
5. Assignment #4
6. ADDITIONAL CASE STUDIES
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The objective of this chapter

- Describe data and process modeling concepts and tools, including data flow diagrams, a data dictionary, and process descriptions.
- Describe the symbols used in data flow diagrams and explain the rules for their use
- Describe various scheduling tools, including Gantt charts and PERT/CPM charts
- Draw data flow diagrams in a sequence, from general to specific
- Explain how to level and balance a set of data flow diagrams
- Describe how a data dictionary is used and what it contains management
- Use process description tools, including structured English, decision tables, and decision trees
- Describe the relationship between logical and physical models

Guidelines for Drawing DFDs

When you draw a context diagram and other DFDs, you should follow several guidelines:

- Draw the context diagram so it fits on one page.
- Use the name of the information system as the process name in the context diagram. For example, the process name in Figure 5-12 is GRADING SYSTEM. Notice that the process name is the same as the system name. This is because context diagram shows the entire information system as if it were a single process. For processes in lower-level DFDs, you would use a verb followed by a descriptive noun, such as ESTABLISH GRADEBOOK, ASSIGN FINAL GRADE,or PRODUCE GRADE REPORT.

Guidelines for Drawing DFDs-continue

When you draw a context diagram and other DFDs, you should follow several guidelines:

- **Use unique names** within each set of symbols. For instance, the diagram in Figure 5-12 shows only one entity named STUDENT and only one data flow named FINAL GRADE. Whenever you see the entity STUDENT on any other DFD in the grading system, you know that you are dealing with the same entity. Whenever the FINAL GRADE data flow appears, you know that you are dealing with the same data flow. The naming convention also applies to data stores
- **Do not cross lines**.

Do not cross lines!!!!

1. One way to achieve that goal is to restrict the number of symbols in any DFD. On lower-level diagrams with multiple processes, you should not have more than nine process symbols. Including more than nine symbols usually is a signal that your diagram is too complex and that you should reconsider your analysis.
2. Another way to avoid crossing lines is to duplicate an entity or data store. When duplicating a symbol on a diagram, make sure to document the duplication to avoid possible confusion. A special notation, such as an asterisk, next to the symbol name and inside the duplicated symbols signifies that they are duplicated on the diagram.

Guidelines for Drawing DFDs-continue

- Provide a unique name and reference number for each process. Because it is the highest-level DFD, the context diagram contains process 0, which represents the entire information system, but does not show the internal workings. To describe the next level of detail inside process 0, you must create a DFD named diagram 0, which will reveal additional processes that must be named and numbered. As you continue to create lower-level DFDs, you assign unique names and reference numbers to all processes, until you complete the logical model.
- Obtain as much user input and feedback as possible. Your main objective is to ensure that the model is accurate, easy to understand, and meets the needs of its users.

CONTEXT DIAGRAM FOR AN ORDER SYSTEM

EXAMPLE: CONTEXT DIAGRAM FOR AN ORDER SYSTEM The context diagram for an order system is shown in Figure 5-13. Notice that the ORDER SYSTEM process is at the center of the diagram and five entities surround the process. Three of the entities, SALES REP, BANK, and ACCOUNTING, have single incoming data flows for FIGURE 5-13

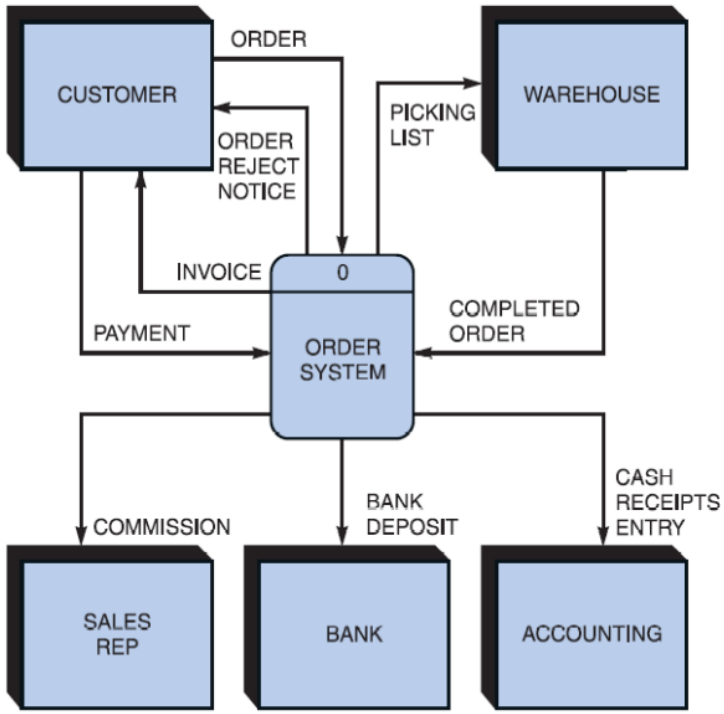


FIGURE 5-13 Context diagram DFD for an order system.

Context Diagram for grading system

that the model is accurate, easy to understand, and meets the needs of its users.

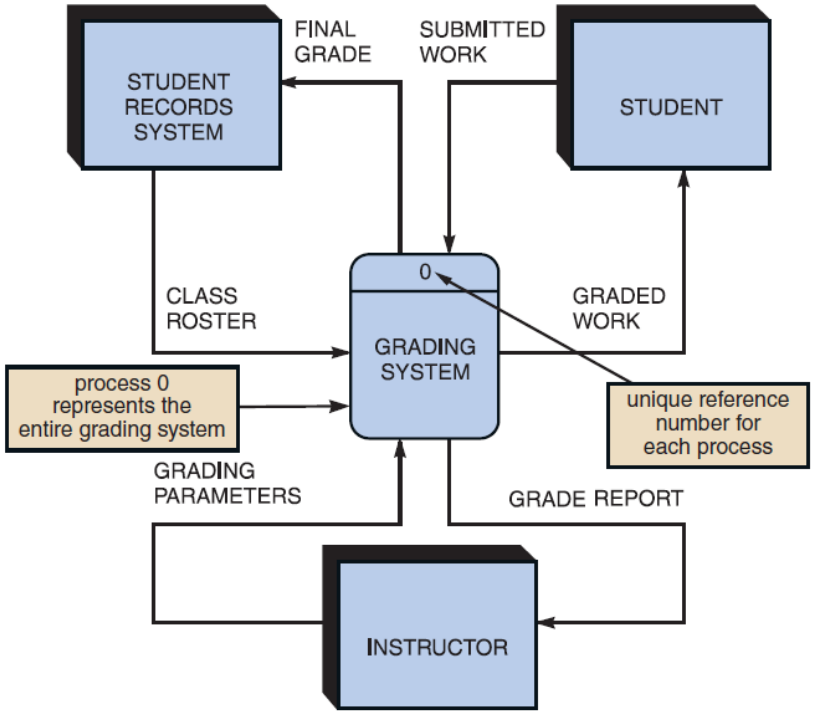
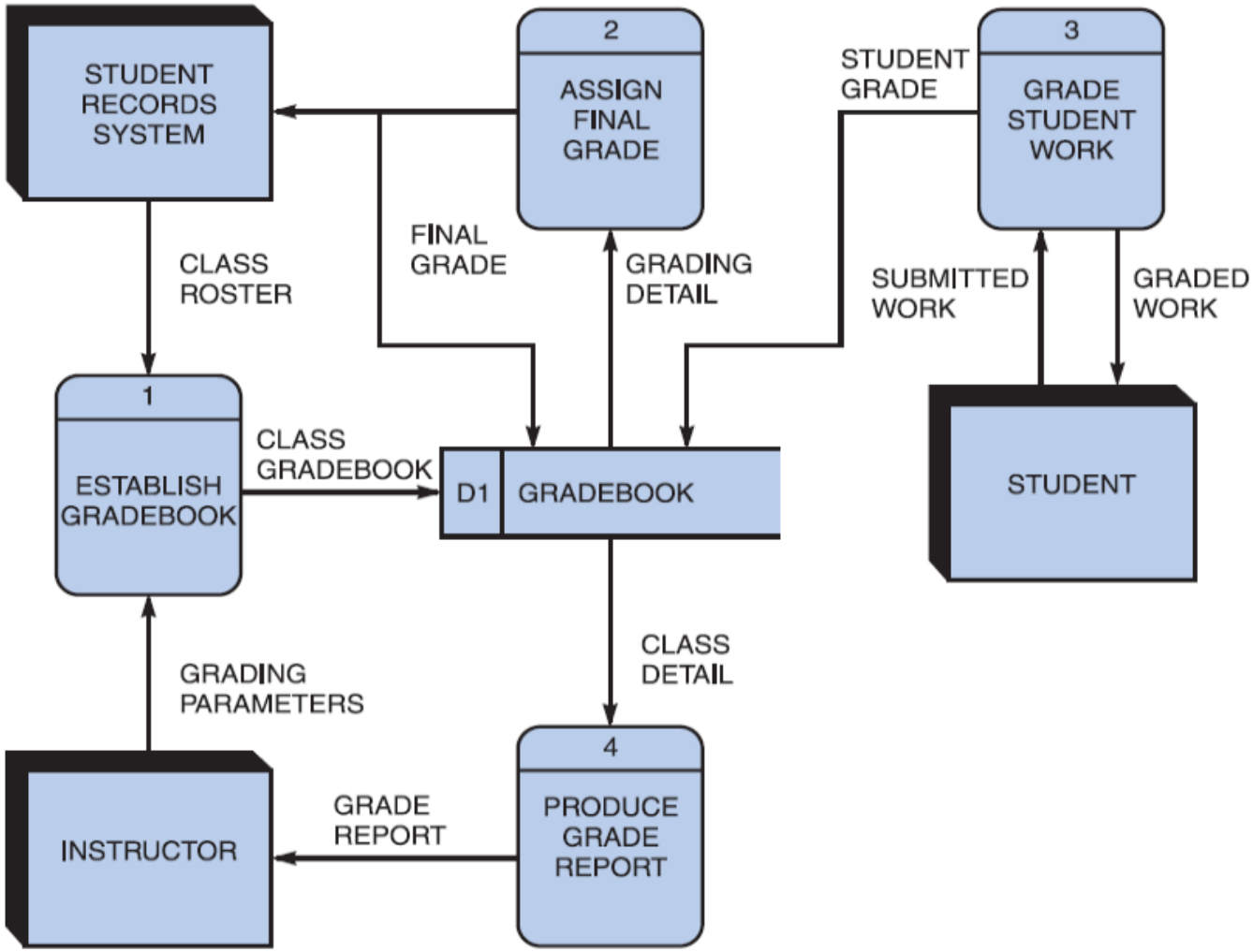


FIGURE 5-12 Context diagram DFD for a grading system.

Figure: Caption

DFD OR DIAGRAM 0 FOR GRADING SYSTEM



DFD OR DIAGRAM 0 FOR ORDER SYSTEM

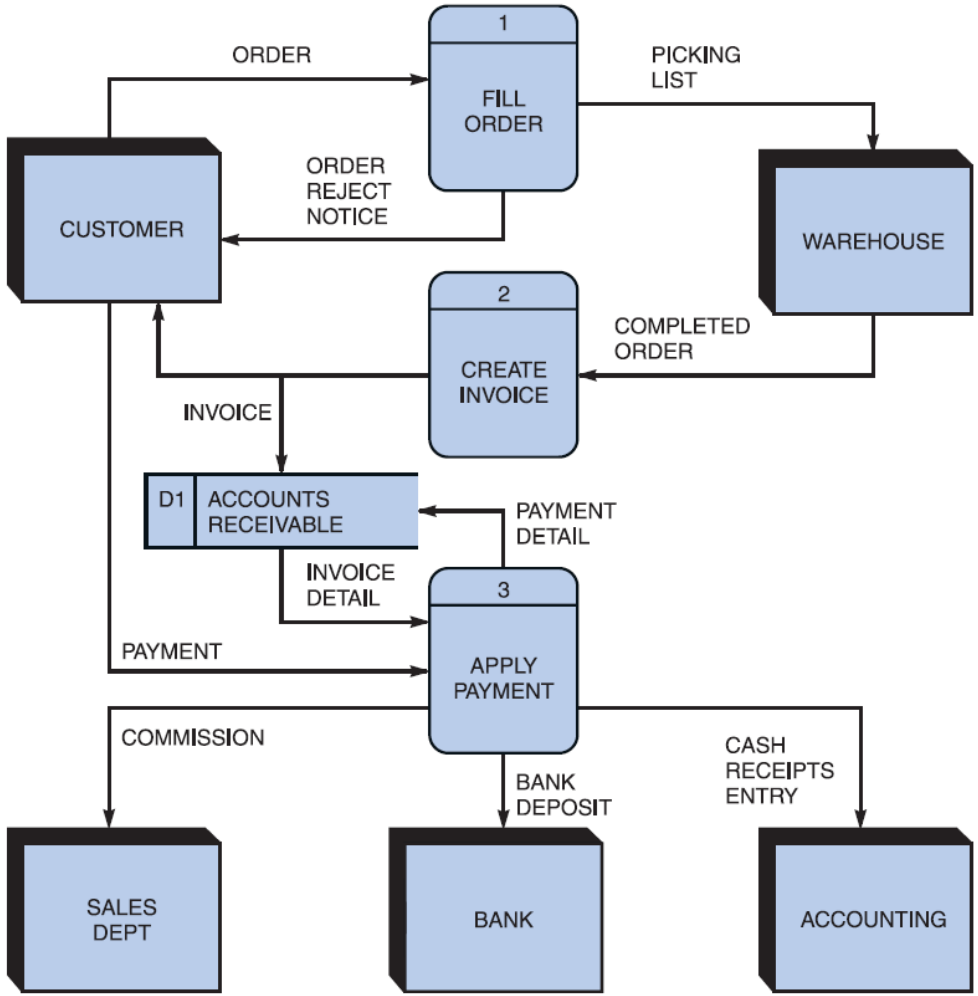


FIGURE 5-16 Diagram 0 DFD for the order system.

Figure: Caption

DATA DICTIONARY

- A set of DFDs produces a logical model of the system, but the details within those DFDs are documented separately in a data dictionary, which is the second component of structured analysis
- A data dictionary, or data repository, is a central storehouse of information about the system's data.
- An analyst uses the data dictionary to collect, document, and organize specific facts about the system, including the contents of data flows, data stores, entities, and processes.
- **The data dictionary** also defines and describes all data elements and meaningful combinations of data elements.
- **A data element**, also called a data item or field, is the smallest piece of data that has meaning within an information system.

DATA DICTIONARY

- **A data element**, also called a data item or field, is the smallest piece of data that has meaning within an information system.
- Examples of data elements are student grade, salary, Social Security number, account balance, and company name.
- Data elements are combined into records, also called data structures.
- A record is a meaningful combination of related data elements that is included in a data flow or retained in a data store.
- For example, an auto parts store inventory record might include part number, description, supplier code, minimum and maximum stock levels, cost, and list price.

Data dictionary

1. Online or manual documentation entries often indicate which system is involved. This is not necessary with a CASE tool because all information is stored in one file that is named for the system.
2. The data element has a standard label that provides consistency throughout the data dictionary.
3. The data element can have an alternative name, or alias.
4. This entry indicates that the data element consists of nine numeric characters.
5. Depending on the data element, strict limits might be placed on acceptable values.
6. The data comes from the employee's job application.
7. This entry indicates that only the payroll department has authority to update or change this data.
8. This entry indicates the individual or department responsible for entering and changing data.

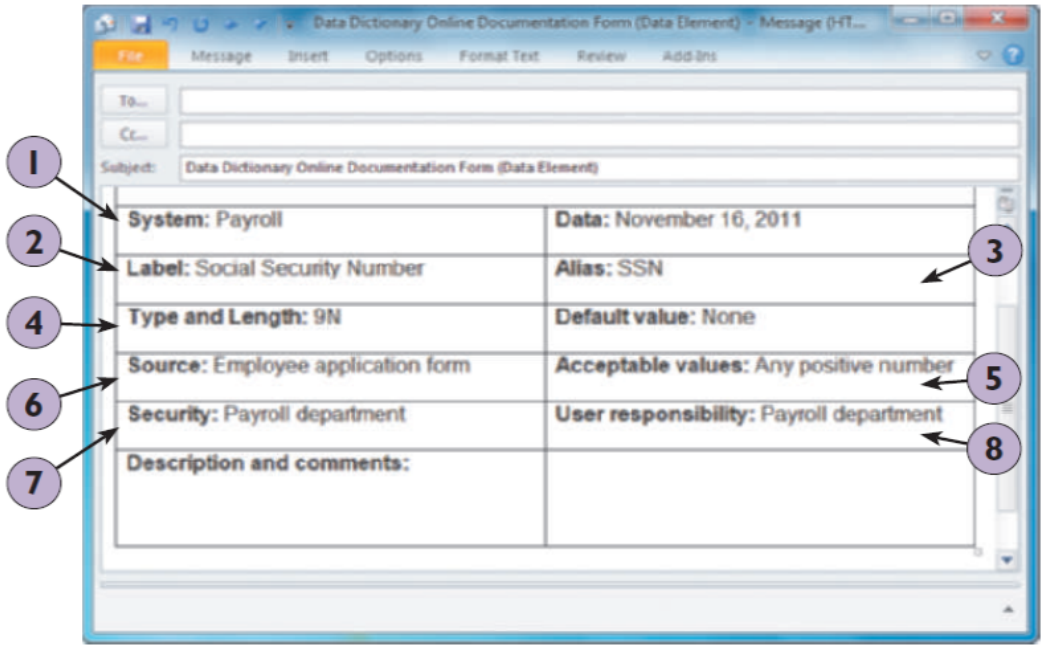


FIGURE 5-23 Using an online documentation form, the analyst has recorded information for a data element named SOCIAL SECURITY NUMBER. Later, the analyst will create a data dictionary entry using a CASE tool.

Figure: Caption

Summery one

- During data and process modeling, a systems analyst develops graphical models to show how the system transforms data into useful information. The end product of data and process modeling is a logical model that will support business operations and meet user needs. Data and process modeling involves three main tools: data flow diagrams, a data dictionary, and process descriptions.
- Data flow diagrams (DFDs) graphically show the movement and transformation of data in the information system. DFDs use four symbols: The process symbol transforms data; the data flow symbol shows data movement; the data store symbol shows data at rest; and the external entity symbol represents someone or something connected to the information system. Various rules and techniques are used to name, number, arrange, and annotate the set of DFDs to make them consistent and understandable.

Summery Continue

- A set of DFDs is like a pyramid with the context diagram at the top.
- The context diagram represents the information system's scope and its external connections but not its internal workings.
- Diagram 0 displays the information system's major processes, data stores, and data flows and is the exploded version of the context diagram's process symbol, which represents the entire information system.
- Lower-level DFDs show additional detail of the information system through the leveling technique of numbering and partitioning.
- Leveling continues until you reach the functional primitive processes, which are not decomposed further and are documented with process descriptions. All diagrams must be balanced to ensure their consistency and accuracy.

Summery 3

- The data dictionary is the central documentation tool for structured analysis. All data elements, data flows, data stores, processes, entities, and records are documented in the data dictionary. Consolidating documentation in one location allows you to verify the information system's accuracy and consistency more easily and generate a variety of useful reports.

Assignment No 4

Answer the following question and submitted the answer before next lecture

1. Describe data and process modeling, and name the main data and process modeling techniques.
2. Describe the Gane and Sarson symbols used for processes, data flows, data stores, and entities. Give four examples of typical names for processes, data flows, data stores, and entities.
3. Describe a data dictionary and give examples of how and when it is used.

Other CaseS studies

1. Draw a context diagram and a diagram 0 DFD that represent the registration system at your school or an imaginary school.
2. On the Internet, locate at least three firms that offer CASE tools. Write e-mail messages to the companies to find out whether they offer demonstration copies or student versions of their products.
3. Suppose that you want to demonstrate a decision table to someone who has never seen one. Think of an example, with two or three conditions, from everyday life. Draw a decision table that captures all possible outcomes.
4. The data flow symbols shown on page 201 were designed by Ed Yourdon, a wellknown IT author, lecturer, and consultant. Many IT professionals consider him to be among the most influential men and women in the software field. Learn more about Mr. Yourdon by visiting his Web site at www.yourdon.com, and write a brief review of his accomplishments.

QA

